

1 REFLX: REACTION ENHANCEMENT IN FITNESS
2 USING LIGHT-BASED EXERCISES FOR UPV ATHLETES

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certifies that this is the approved version of the following special problem:

**REFLX: REACTION ENHANCEMENT IN FITNESS
USING LIGHT-BASED EXERCISES FOR UPV ATHLETES**

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Dedication

“Hello, world.”

Acknowledgment

“Hello, world.”

Abstract

From 150 to 200 words of short, direct and complete sentences, the abstract should be informative enough to serve as a substitute for reading the entire SP document itself. It states the rationale and the objectives of the research. In the final Special Problem document (i.e., the document you'll submit for your final defense), the abstract should also contain a description of your research results, findings, and contribution(s).

Suggested keywords based on ACM Computing Classification system can be found at https://dl.acm.org/ccs/ccs_flat.cfm

Keywords: Keyword 1, keyword 2, keyword 3, keyword 4, etc.

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Chapter 1

Introduction

1.1 Overview of the Current State of Technology

In the realm of sports science, the enhancement of athletes' response time and agility remains a critical focus, as these attributes directly influence performance in dynamic, unpredictable environments such as team sports and combat disciplines (Hassan et al., 2023). Traditional training methods, including cone drills and ladder exercises, have long been employed to improve these skills, yet they often fall short in replicating the rapid, stimulus-driven demands of real-game scenarios. Over the past decade, light-based reaction training systems—devices utilizing visual stimuli like LED lights to prompt immediate motor responses—have emerged as innovative tools to bridge this gap. Systems such as FITLIGHT, BlazePod, and XLiGHT have been critically analyzed for their design features, including sensor connectivity, battery life, and operational reliability, revealing strengths in portability and customization but limitations in diagnostic precision

91 and validity (Ezhov et al., 2021).

92 Empirical studies have demonstrated that these systems can significantly enhance
93 visual-motor coordination, reaction speed, and cognitive functions. For instance,
94 interventions using FITLIGHT in small-sided games have led to marked improve-
95 ments in harmonic abilities (e.g., rhythmization and responsiveness) and basic
96 skills like dribbling among young basketball players (Hassan et al., 2023). Simi-
97 larly, a 10-week FITLIGHT program improved reaction times and dribbling speeds
98 in female basketball athletes, with effect sizes indicating substantial neural adapta-
99 tions (Hassan, 2025). In motorsport contexts, light-based reactive agility training
100 has boosted selective attention, cognitive flexibility, and cardiorespiratory capac-
101 ity in car racing drivers (Horváth et al., 2022). A systematic review of visual
102 training interventions, including light board and stroboscopic methods, further
103 corroborates these benefits, reporting 5-27% reductions in reaction time across
104 various sports, with greater efficacy in elite and younger athletes (Jothi et al.,
105 2025). Reliability assessments of systems like BlazePod have also affirmed their
106 validity for measuring simple and complex reactions in mixed martial arts (MMA)
107 athletes, with high intraclass correlations supporting their use in training proto-
108 cols (Polechoński et al., 2024).

109 Despite these advancements, significant gaps persist in the literature. While light-
110 based systems show promise in controlled settings, their predictive value for field-
111 based reactive agility remains limited, as evidenced by weak correlations between
112 laboratory reaction speeds and on-field performance in soccer players (Broodryk
113 et al., 2025). This suggests a disconnect between isolated visual stimuli and the
114 multifaceted perceptual-cognitive demands of sports, highlighting the need for
115 more integrated, sport-specific designs. Moreover, comparative analyses under-

score inconsistencies in system performance, such as variable Bluetooth stability and sensor delays, which could undermine training reproducibility (Ezhov et al., 2021). These issues are largely tied to the wireless protocols used by commercial systems—typically Bluetooth or Wi-Fi—both of which introduce transmission overhead and variable latency, limiting their suitability for real-time reaction measurement. This technological gap highlights the need to explore emerging low-latency communication protocols.

1.2 Problem Statement

While traditional methods such as cone and ladder drills have been widely used to enhance reaction time and agility, they often fail to simulate the complex, stimulus-driven conditions of real-game scenarios (Hassan et al., 2023). Recent light-based reaction training systems have shown potential in improving visual-motor coordination and cognitive response (Horváth et al., 2022; Jothi et al., 2025). However, these systems remain limited by high costs, connectivity issues, and questionable transferability of laboratory-based improvements to on-field performance (Broodryk et al., 2025; Ezhov et al., 2021).

To address these limitations, this study bridges engineering innovation and applied sports performance research by incorporating ESP-NOW—a low-latency, connectionless wireless communication protocol developed by Espressif—in its development of a low-cost and customizable light-based reaction training system. By leveraging ESP-NOW’s novel capabilities, the system aims to eliminate the communication bottlenecks present in existing technologies and provide a more

temporally accurate, cost-efficient, and reliable training tool. If such limitations in current systems persist, athletes and coaches will continue to rely on devices that cannot fully simulate realistic gameplay conditions nor guarantee measurement precision. Thus, the integration and evaluation of ESP-NOW become central to developing a responsive and scientifically validated reaction training system.

1.3 Research Objectives

1.3.1 General Objective

The goal of this study is to develop and evaluate a light-based reaction training system that enhances the response time and agility of athletes, specifically in racket sports such as badminton, tennis, and table tennis.

1.3.2 Specific Objectives

Specifically, this study aims to:

1. design and construct a device equipped with infrared sensors, RGB lights, and speakers while integrating ESP-NOW for accurate motion detection and response measurement,
2. develop a software application that manages the device operations, records performance data, and functions both online and offline,

- 155 3. calibrate and test the system to ensure precision, responsiveness, and syn-
156 chronization between hardware and software components.
- 157 4. conduct experimental trials assessing the system's effectiveness in improving
158 athletes' response time and agility compared to traditional training methods,
159 and
- 160 5. evaluate the system's usability, functionality, and overall user satisfaction
161 based on feedback from athletes and coaches.

162 1.4 Scope and Limitations of the Research

163 This study focuses on the design, development, and short-term evaluation of a
164 programmable light-based reaction training system specifically tailored for racket
165 sports, including badminton, tennis, and table tennis. These sports were selected
166 because they demand rapid visual processing, anticipatory decision-making, and
167 fine motor control—abilities strongly linked to reaction time and agility.

168 Experimental trials are conducted in controlled indoor training environments, us-
169 ing drills that simulate racket-sport scenarios such as serve returns, directional
170 changes, and split-step reactions. The prototype system incorporates LED visual
171 cues and integrated speakers to deliver multimodal stimuli, allowing assessment
172 of both single and dual sensory response conditions.

173 The evaluation is limited to short-term performance outcomes, measuring pre-
174 post changes in reaction and movement response metrics following exposure to the
175 prototype system. Long-term effects, such as learning retention, in-game transfer,

176 or perceptual-cognitive adaptations, are beyond the scope of this research.

177 1.5 Significance of the Research

178 This research integrates engineering, computer science, and sports science by de-
179 veloping a light- and sound-based reaction training system that enhances athletes'
180 response time and agility through multimodal stimuli and data-driven feedback.
181 Through a technical lens, the study contributes to the computer science commu-
182 nity by implementing real-time sensor processing, audio-visual cue synchroniza-
183 tion, and a user-centered software design. The inclusion of both infrared sensors
184 and speakers allows for dynamic and varied training scenarios that engage multi-
185 ple sensory pathways, thereby improving cognitive-motor coordination. The use
186 of ESP-NOW as the primary wireless protocol for synchronizing training stations
187 is the central innovation this study. Unlike traditional Bluetooth-based systems,
188 ESP-NOW enables stable, connectionless, and ultra-low-latency communication,
189 thereby ensuring accurate temporal measurements essential for reaction-time re-
190 search. Its novel application into a light-based athletic training device positions
191 this study at the intersection of sports technology and modern IoT communication
192 design.

193 From a societal perspective, the system democratizes access to advanced reaction
194 training technologies by providing an affordable and portable tool suitable for ath-
195 letes, coaches, and educational institutions. It can also serve as a supplementary
196 device for rehabilitation programs that aim to improve motor control and sensory
197 processing. By combining technical innovation with accessibility, this research

198 promotes evidence-based athletic development and supports the broader integra-
199 tion of smart, adaptive technologies in sports and human performance training.

Chapter 2

Review of Related Literature

2.1 Physiological Basis of Reaction and Agility

Reaction time and agility in athletes are influenced by a complex interplay of neurophysiological factors, including perceptual-cognitive processing, neural pathways, and motor control. Reaction time encompasses the interval from stimulus detection to response initiation, modulated by sensory input (primarily visual), central nervous system processing, and efferent motor commands. Agility, defined as the ability to change direction rapidly while maintaining balance and speed, integrates these with biomechanical elements like strength and coordination (Pojskic et al., 2019). Neurophysiological mechanisms involve the visual cortex for stimulus detection, the prefrontal cortex for decision-making, and the basal ganglia for motor planning, with reaction accuracy linked to efficient attentional orienting and split attention across multiple stimuli (Chow et al., 2022). The prefrontal studies highlight that perceptual factors, such as anticipation and visual search efficiency,

215 contribute significantly, alongside physical attributes like explosive strength and
216 elastic strength (Yildiz et al., 2020). For instance, factor analyses reveal indepen-
217 dent components: explosive strength for acceleration, elastic strength for rebound,
218 change-of-direction speed (CODS), and maximal strength. Correlations show that
219 faster reaction times predict superior agility test performances, such as in the Illi-
220 nois Agility Test or 20-m shuttle sprint, emphasizing the role of neural efficiency
221 (Wang et al., 2024).

222 Methodologies in these studies often employ systematic reviews and correlational
223 analyses, drawing from diverse athletic populations (e.g., team sports like soc-
224 cer) (Turna, 2020). Limitations include reliance on lab-based tests that may not
225 fully replicate field conditions, potentially underestimating contextual factors like
226 fatigue or cognitive load (Pojskic et al., 2019). Implications suggest that train-
227 ing targeting neuroplasticity—through repeated stimuli—can enhance synaptic
228 efficiency, reducing reaction latencies and improving agility in sports requiring
229 rapid responses, such as basketball or racing (Chow et al., 2022). Patterns indi-
230 cate stronger associations in high-level athletes, where cognitive fatigue and sleep
231 quality positively correlate with prolonged reaction times, highlighting the need
232 for holistic training approaches (Yildiz et al., 2020).

233 2.2 Existing Reaction Training Technologies

234 Commercial systems like FITLIGHT Trainer and BlazePod represent established
235 light-based technologies for reaction training, with evidence supporting their ef-
236 fectiveness in enhancing athletic performance. FITLIGHT, a wireless LED sensor

237 system, simulates game-like conditions to improve reaction time, reflexes, and
238 cognitive functions (Hassan, 2025). Studies demonstrate its reliability, with a
239 test-retest intraclass correlation coefficient (ICC) of 0.89 and minimal detectable
240 changes in reaction metrics (Myers, 2021). In young basketball players, a 10-week
241 FITLIGHT intervention improved executive functions (e.g., inhibition, working
242 memory) and fitness, though gains were comparable to traditional training, sug-
243 gesting added cognitive demand without superior outcomes (Hassan, 2025). An-
244 other trial integrated FITLIGHT into small-sided games for 18 weeks, yielding
245 significant enhancements in coordinative abilities and basic skills, outperforming
246 controls with large effect sizes (Steff et al., 2024).

247 BlazePod, a pod-based visual-cognitive system, shows similar promise, with ex-
248 cellent reliability in balance activities (Çekok & Anaforoğlu, 2025). A 6-month
249 program in adolescent soccer players improved simple reaction time and cognitive
250 tasks, but between-group differences were non-significant compared to standard
251 training, indicating contextual benefits rather than inherent superiority (Theofilou
252 et al., 2022). Methodologies typically use randomized controlled trials (RCTs)
253 with pre-post assessments, validated via tools like the Stroop Test for cognition
254 and TUG for effort (Çekok & Anaforoğlu, 2025). Limitations include fairly small
255 samples ($n = 20-70$) and short durations (6-18 weeks), risking overestimation
256 of effects due to novelty; discrepancies arise in massed vs. distributed schedul-
257 ing, with longer protocols showing sustained gains. Implications underscore these
258 systems' role in individualized drills, boosting engagement through gamification,
259 though high costs and proprietary designs limit accessibility for broader athletic
260 populations (Hassan, 2025).

2.3 Light- and Sensor-Based Training Research

Research on light-based stimuli consistently shows enhancements in sports performance, particularly cognitive-motor integration and agility. A 6-week RCT using Witty SEM lights for car racing drivers improved cognitive abilities and cardiorespiratory fitness, with significant group-time interactions (Horváth et al., 2022). In basketball, light stimulation exercises enhanced attention focus (visual/auditory) and skilled hand speed, with pre-post improvements attributed to neuroplastic adaptations (Shimi et al., 2025). VR-adapted light tasks revealed attentional mechanisms, with orienting speed predicting performance, emphasizing split attention for larger stimuli arrays (Shimi et al., 2025). Soccer-specific lighting interventions (6 months) manipulated visual processing, reducing reaction times under varied conditions, though transfer to matches was inferred rather than directly measured (Theofilou et al., 2022).

Methodologies favor RCTs with tools like the Vienna Test System for cognition and breath-by-breath gas analysis for physiology, ensuring objective evidence (Horváth et al., 2022). Limitations encompass single-blinding, uncontrolled nutrition, and lab-based focus, potentially inflating effects; patterns show greater benefits in open-skill sports, with discrepancies in exercise vs. rest conditions (Zhao et al., 2024). Implications highlight light stimuli's potential for dual physical-cognitive gains, transferable to agility-dependent sports, but call for ecological validity in future studies (Shimi et al., 2025).

2.4 Wireless Communication Protocols in IoT

The rapid evolution of Internet of Things (IoT) and embedded-device systems has introduced novel communication protocols designed for low-latency, low-power, peer-to-peer connectivity. Among these, ESP-NOW stands out as a connectionless, peer-to-peer protocol by Espressif that operates on the Wi-Fi PHY layer while bypassing the full TCP/IP stack, which reduces handshake overhead and enables low-latency broadcast and one-to-many messaging for ESP32 and ESP8266 devices (Becker et al., 2025). Multiple empirical evaluations have assessed ESP-NOW's performance under different environmental and traffic conditions.

An outdoor performance study reported one-way transmission delays as low as approximately 1–2 ms even at extended distances, showing that ESP-NOW maintains stable latency and packet delivery across open-field environments (Becker et al., 2025). A lab analysis further confirmed these results by comparing ESP-NOW directly with Bluetooth Low Energy (BLE), finding that ESP-NOW exhibited measurably faster transmission times, significantly lower jitter, and more stable multi-node communication—attributes essential for synchronized sensor systems (Labib et al., 2021). Similarly, a comparative evaluation showed that ESP-NOW outperformed both classic Bluetooth and standard Wi-Fi in key metrics such as transmission speed, barrier resistance, and latency, including tests through walls and indoor obstacles (Eridani et al., 2021). An indoor performance evaluation of ESP-NOW reported high packet-delivery ratios and consistent millisecond-level timing across typical indoor IoT deployment ranges (Urazayev et al., 2023). These studies show that ESP-NOW reliably achieves one-way transmission times in the range of about 1–5 ms with low jitter—performance characteristics.

2.5 Chapter Summary

Should include a table of related studies comparing them based on several criteria.

Highlight research gaps and the research problem.

The literature reveals notable gaps, particularly the scarcity of open-source, customizable systems for individualized athlete training. While commercial tools like FITLIGHT dominate, they are proprietary and costly, limiting adaptation for diverse needs (Seçkin et al., 2023). Emerging technologies emphasize open-source platforms for tele-exercise, enabling collaborative customization via AI and IoT, yet only 9% of trials target healthy populations, with inadequate standardization of training parameters (Rebelo et al., 2023). Gaps include high dropout in asynchronous modes, data security concerns, and underexplored adherence factors (Rebelo et al., 2023). Methodologies in gap analyses rely on narrative reviews, highlighting sustainability challenges in open-source projects (Seçkin et al., 2023). This implicates the need for accessible systems to democratize training, addressing disparities in research on healthy athletes.

Another notable gap is the absence of research exploring low-latency wireless communication protocols—particularly ESP-NOW—in the design of reaction-training technologies. Existing systems largely depend on Bluetooth, which suffers from variable transmission delays and connectivity interruptions. No current studies assess whether protocols like ESP-NOW can enhance timing precision or improve synchronization in multi-station athletic training systems, underscoring a significant technological gap this research intends to address.

Chapter 3

Research Methodology

This study follows a step-by-step research process to develop and evaluate a light-based reaction training system for racket sports. The methodology includes gathering background information, consulting experts, and creating the initial system design. It then covers building the hardware prototype, developing the software, and testing the system with UPV athletes to measure reaction time and agility. Each stage is planned to ensure that the device is accurate, usable, and aligned with actual training needs. The process ends with analyzing the results and completing the final technical documentation.

3.1 Research Activities

Research activities include inquiry, survey, research, brainstorming, canvassing, consultation, review, interview, observe, experiment, design, test, document, etc.

3.1.1 Research and Consultation

The researchers consulted with select UPV PE Faculty to ensure and validate the feasibility and impact of the proposed device for racket sports, with said consultations taking place at the UPV Covered Court. A comprehensive research on reaction time training that utilizes light-based systems, especially those for racket sports was done during this stage of the methodology. This stage helped to establish a strong theoretical foundation, identify gaps, and ensure that the design aligns with athletic performance improvement.

3.1.2 Brainstorming and Prototype Development

A number of UPV Computer Science and PE Faculty assisted the researchers in developing the final design concept of the system, which includes the integration of infrared (IR) sensors, LED light modules, and speakers. The procurement of materials and parts was done during this phase and afterwards a prototype was developed for initial testing.

Figure 3.1 illustrates the schematic circuit diagram of the master controller. At the core of this design is a standard ESP32 microcontroller, which manages two primary tasks: sending commands to the station modules and collecting the reaction-time data they generate. The main controller also includes a DF Mini MP3 player module that provides audio cues for training modes requiring sound-based stimuli by reading audio files stored on an SD card and outputting them through a connected 3W speaker for clear and amplified playback. The entire system is powered by a 3.7V 18650 battery and a TP4056 charging module with built-in protection

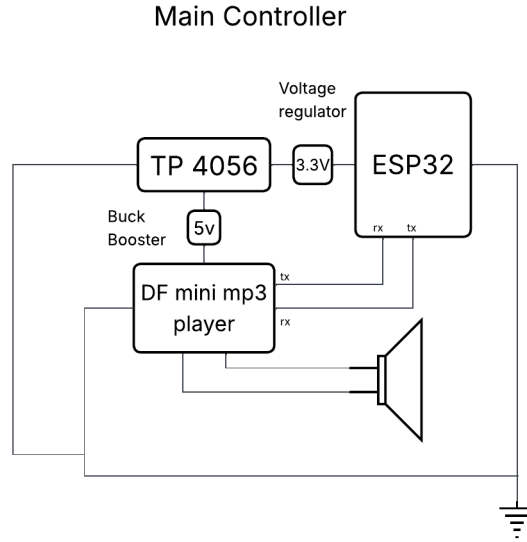


Figure 3.1: Schematic Circuit Diagram of the Master Controller

363 ensures that the battery safely charges. Lastly, the A 3.3V voltage regulator pro-
 364 vides a stable supply for the ESP32, while a separate step-up converter generates
 365 the required 5V output for the DF Mini MP3 module and the speaker.

366 Figure 3.2 illustrates the schematic circuit diagram of the station controllers. Each
 367 station uses an ESP32 Super Mini C3 microcontroller, chosen for its compact size
 368 and sufficient number of GPIO pins to support the required sensors. The station
 369 node receives commands from the main controller and executes them accordingly.
 370 It controls a programmable LED strip, which lights up based on the received com-
 371 mand and turns off when the connected infrared sensor detects a response. The
 372 station then records the performer's reaction time and sends this data back to the
 373 main controller. Like the main controller, each station node is powered by a 3.7V
 374 18650 battery paired with a TP4056 charging module with built-in protection.
 375 A voltage regulator provides a stable 3.3V supply for the microcontroller, while
 376 a step-up converter generates the 5V needed for the LED strip. The LED strip

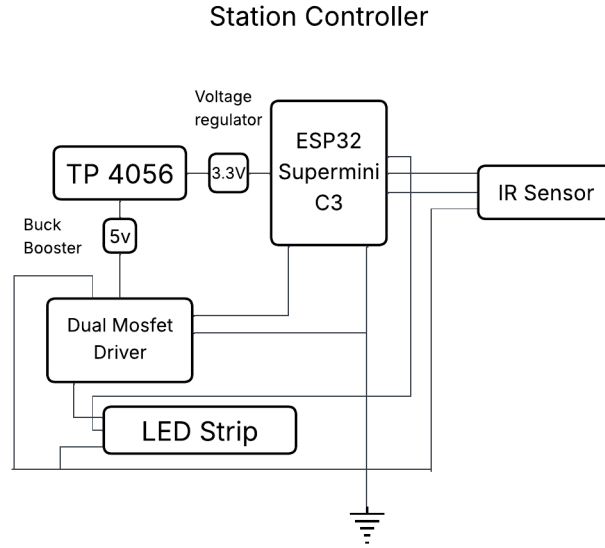


Figure 3.2: Schematic Circuit Diagram of the Station Controller

377 is connected through a dual MOSFET driver, which serves as a switch to safely
 378 control the 5V load from the station controller.

379 3.1.3 Latency Benchmarking of Bluetooth vs ESP-NOW

380 The wireless communication latency of the ESP-NOW protocol and Bluetooth was
 381 empirically benchmarked under controlled indoor conditions. Since reaction-time
 382 assessment is highly sensitive to communication delays, quantifying the trans-
 383 mission latency between the master controller and station nodes is essential for
 384 validating the system's responsiveness and synchronization reliability.

385 The benchmarking setup consisted of one master controller and one station con-
 386 troller positioned at varying distances of approximately 1 meter, 2 meters, and
 387 5 meters within an indoor environment similar to the intended deployment set-
 388 ting. Each station controller was configured as an ESP-NOW peer of the master

389 controller, as well as all devices operating on a fixed Wi-Fi channel to minimize
390 external interference.

391 Latency measurements were obtained using a timestamp-based approach. Upon
392 initiating a stimulus command, the master controller records a transmission times-
393 tamp using the ESP32's internal clock. Upon reception of the command, the
394 station controller immediately transmits an acknowledgment packet back to the
395 master controller. The master controller then records the reception timestamp of
396 this acknowledgment. One-way latency was estimated from the transmission-to-
397 reception interval at the station, while round-trip latency was computed as the
398 time difference between command transmission and acknowledgment reception at
399 the master controller.

400 Each latency test was repeated over 50 trials per distance (1 meter, 2 meters, and
401 5 meters) to account for variability. The mean latency, standard deviation, and
402 worst-case delay was computed for each distance condition. The results of this
403 benchmark was used as the empirical basis for selecting ESP-NOW as the sys-
404 tem's wireless protocol, demonstrating that its low-latency, connectionless design
405 is suitable for synchronized, multi-station reaction training where precise timing
406 is critical.

407 **Chapter 4**

408 **Results and Discussions**

409 This chapter presents the results or the system of your SP. Include screenshots,
410 tables, or graphs and provide the discussion of results.

411 **Chapter 5**

412 **Conclusion**

413 This chapter summarizes your SP and provides conclusions regarding your results
414 and analyses. Provide recommendations on what ought to be done with your SP
415 or provide further directions on the topic you covered.

Chapter 6

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509 **Appendix A**

510 **Code Snippets**

511 Appendix B

512 Resource Persons

513 **Dr. Firstname1 Lastname1**

514 Role1

515 Affiliation1

516 emailaddr@domain.com

517 **Mr. Firstname2 Lastname2**

518 Role2

519 Affiliation2

520 emailaddr2@domain.com

521 **Ms. Firstname3 Lastname3**

522 Role3

523 Affiliation3

524 emailaddr3@domain.net

525